

ANN ARBOR, MI, USA

WMO: 725374

Lat: 42.223N

Lon: 83.744W

Elev: 256

StdP: 98.29

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 94889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-15.8	-22.7	0.5	-18.3	-19.3	0.7	-14.3	11.2	-4.2	10.2	-4.4	2.1	230	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	32.1	23.0	30.6	22.5	28.9	21.5	24.7	29.6	23.7	28.3	22.8	27.2	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.3	27.2	22.3	17.5	26.4	21.3	16.5	25.3	75.8	29.5	71.9	28.3	68.4	27.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.5	7.6	-24.5	34.0	3.7	1.7	-27.1	35.2	-29.3	36.1	-31.3	37.1	-34.0	38.3		
(5)			-24.6	26.4	3.8	1.0	-27.3	27.1	-29.5	27.7	-31.6	28.2	-34.4	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.7	-3.8	1.8	8.7	14.6	19.4	21.6	20.7	16.8	10.2	4.0	-1.6
	DBStd	10.49	6.39	6.07	5.93	4.88	4.54	3.52	3.22	2.87	4.21	4.83	5.04	5.57
	HDD10.0	1806	457	387	262	79	11	0	0	0	2	58	188	361
	HDD18.3	3697	714	620	512	289	133	29	8	10	77	256	429	618
	CDD10.0	1461	1	1	9	42	154	282	359	332	208	64	9	2
	CDD18.3	309	0	0	0	2	18	61	109	84	32	4	0	0
	CDH23.3	3251	0	0	6	32	226	654	1174	796	323	41	0	0
(14) Wind	WSAvg	3.4	3.9	3.9	3.9	4.0	3.3	2.9	2.7	2.6	2.6	3.3	3.6	3.8
	PrecAvg	888	55	49	72	91	87	90	83	86	78	60	71	68
	PrecMax	1217	127	158	155	184	198	215	171	222	179	194	149	126
	PrecMin	504	5	4	22	12	27	15	11	7	7	2	19	9
	PrecStd	133	30	31	30	35	41	40	43	53	38	33	32	30
	DB	12.7	15.2	22.9	27.1	30.3	32.9	34.7	32.8	32.1	27.5	20.8	15.2	
	MCWB	10.6	11.3	16.4	17.7	21.5	23.0	23.8	23.2	22.5	20.9	14.2	13.3	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	9.4	10.8	17.7	23.5	28.0	31.1	32.3	30.8	29.3	23.9	17.3	11.8	
	2% MCWB	7.9	8.9	12.2	15.3	19.8	22.5	23.4	22.8	21.5	17.6	12.6	9.8	
	5% DB	6.3	7.1	14.1	20.9	26.1	28.9	30.9	28.9	27.1	21.4	14.7	8.4	
	5% MCWB	4.7	4.8	9.9	13.7	18.9	21.4	22.8	21.8	20.0	16.2	11.1	6.6	
	10% DB	3.6	3.9	11.3	17.9	23.4	27.2	28.9	27.5	24.8	18.8	12.3	5.9	
	10% MCWB	2.3	1.9	8.0	12.0	17.5	20.4	21.7	21.0	19.0	14.8	9.4	4.1	
	WB	11.6	12.4	16.9	19.1	23.0	24.8	26.1	25.6	24.3	21.6	16.0	13.6	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	12.5	13.6	22.5	24.8	27.6	30.6	31.6	30.4	29.7	26.4	18.7	15.1	
	2% WB	7.9	8.7	13.5	16.8	21.3	23.5	24.9	24.1	22.9	19.3	13.6	10.1	
	2% MCDB	8.8	10.5	16.5	20.8	26.3	28.9	30.2	27.9	27.1	22.4	16.5	11.3	
	5% WB	4.8	4.8	11.0	14.9	19.9	22.5	23.9	23.1	21.5	17.2	11.6	6.9	
	5% MCDB	6.2	6.8	13.3	19.7	24.6	27.7	28.5	27.1	25.0	19.9	14.0	8.3	
	10% WB	2.4	2.2	7.8	12.8	18.4	21.3	22.9	22.1	20.2	15.3	9.5	4.1	
	10% MCDB	3.6	3.8	10.8	17.3	22.5	25.6	27.3	25.9	23.4	18.2	12.1	5.6	
(35) Mean Daily Temperature Range	MDBR	8.3	9.2	10.3	12.1	12.5	12.6	13.1	12.4	13.0	11.8	9.7	7.5	
	5% DB	9.7	11.8	15.0	17.0	15.5	14.9	14.6	14.0	15.4	15.1	13.5	10.1	
	MCWBR	8.4	9.7	10.4	10.0	8.4	7.5	6.8	6.9	8.1	9.2	10.0	8.7	
	5% WB	9.3	11.4	13.9	14.3	13.4	12.8	12.3	11.6	12.5	12.8	11.8	9.4	
	MCWBR	8.4	9.9	10.4	9.6	8.2	7.3	6.8	6.4	7.1	8.6	9.7	8.6	
(40) Clear-Sky Solar Irradiance	taub	0.297	0.337	0.355	0.401	0.428	0.450	0.441	0.431	0.386	0.353	0.316	0.310	
	taud	2.390	2.252	2.328	2.218	2.195	2.170	2.216	2.232	2.342	2.421	2.482	2.383	
	Ebn at Noon	847	855	887	866	850	829	832	828	844	832	821	793	
	Edh at Noon	79	108	113	136	143	147	139	133	110	91	73	74	
(44) All-Sky Solar Radiation	RadAvg	1.53	2.42	3.53	4.60	5.36	6.04	6.09	5.27	4.27	2.78	1.74	1.22	
	RadStd	0.14	0.31	0.35	0.41	0.47	0.39	0.30	0.21	0.38	0.24	0.23	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	N/A

Nomenclature: See separate page